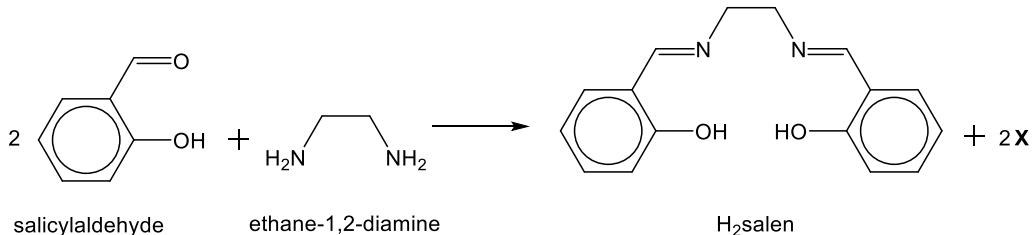
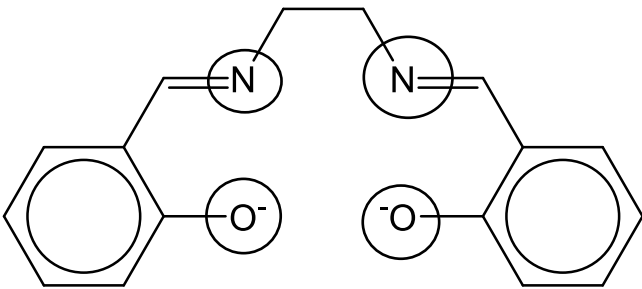
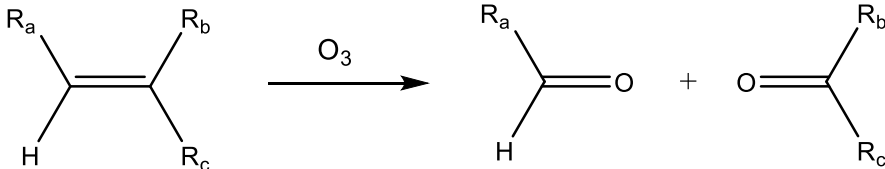
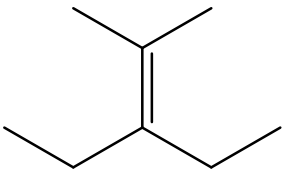
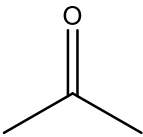
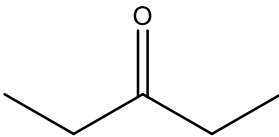
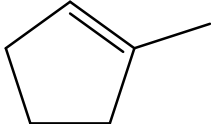
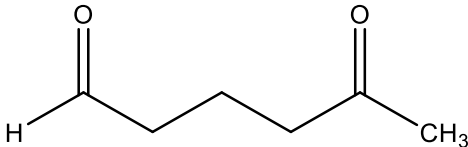


ACJC solutions for H2 Chemistry Prelim Paper 2 2022

1	(a)	Table 1.1 shows the solubility of two organic molecules at 25 °C.																	
		<table><tr><th colspan="2">Table 1.1</th></tr><tr><th></th><th>solubility in water / g dm⁻³</th></tr><tr><td>propanone</td><td>miscible</td></tr><tr><td>chloromethane</td><td>5.04</td></tr></table>			Table 1.1			solubility in water / g dm ⁻³	propanone	miscible	chloromethane	5.04							
Table 1.1																			
	solubility in water / g dm ⁻³																		
propanone	miscible																		
chloromethane	5.04																		
	(i)	Identify the type of intermolecular force present between the molecules. propanone: permanent dipole -permanent dipole chloromethane:.....permanent dipole -permanent dipole[1]																	
	(ii)	Give a reason for the difference in their solubilities in water Propanone is able to form hydrogen bonds with water but chloromethane is unable to.																	
	(b)	The boiling point of three ligands are shown in Table 1.2.																	
		<table><tr><th colspan="3">Table 1.2</th></tr><tr><th>ligand</th><th>formula</th><th>boiling point / °C</th></tr><tr><td>water</td><td>H₂O</td><td>100</td></tr><tr><td>ammonia</td><td>NH₃</td><td>-33.3</td></tr><tr><td>hydrazine</td><td>N₂H₄</td><td>114</td></tr></table>			Table 1.2			ligand	formula	boiling point / °C	water	H ₂ O	100	ammonia	NH ₃	-33.3	hydrazine	N ₂ H ₄	114
Table 1.2																			
ligand	formula	boiling point / °C																	
water	H ₂ O	100																	
ammonia	NH ₃	-33.3																	
hydrazine	N ₂ H ₄	114																	
	(i)	Explain what is meant by the term <i>ligand</i> . A ligand is a neutral molecule or anion species that has a lone pair of electrons to form a dative bond with a central metal atom or ion.																	
	(ii)	Explain the difference in the boiling points of the three ligands. Hydrazine has the highest boiling point as it has the largest/most polarisable electron cloud size has strongest instantaneous dipole-induced dipole. Water form more extensive hydrogen bonds than ammonia and hence it has a higher boiling point.																	
	(c)	Polydentate ligands are ligands which form more than one bond with the metal atom or ion. Salicylaldehyde, ethane-1,2-diamine and H ₂ salen are examples of such ligands. H ₂ salen can be synthesised from salicylaldehyde and ethane-1,2-diamine.																	

		 salicylaldehyde ethane-1,2-diamine H₂salen When the phenolic groups of H₂salen are deprotonated, salen²⁻ acts as a ligand. It has a high affinity for Co²⁺ ions and forms a planar complex Co(salen).
		(i) Suggest the identity of molecule X. H₂O
		(ii) Draw the structure of a salen²⁻ ligand and circle the atoms which are used to coordinate to a Co²⁺ ion.  salen²⁻ [2]
		(iii) State the coordination number of the Co²⁺ ion in Co(salen). Four
		(d) Ozonolysis is a method to oxidatively cleave alkenes using ozone, O₃, to form carbonyl compounds. 

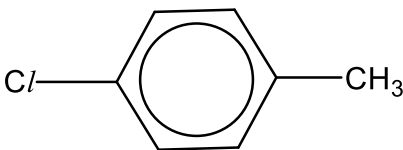
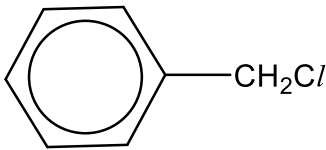
		(i)	 2-methyl-3-ethylpent-2-ene Draw the structure of the organic products when 2-methyl-3-ethylpent-2-ene undergoes ozonolysis.   [2]
		(ii)	Ozonolysis of A, C_6H_{10}, gives a single compound, B, $C_6H_{10}O_2$. B gives a yellow precipitate when treated with alkaline aqueous iodine and forms a red-brown precipitate when treated with Fehling's solution. Draw the structures of the compounds A and B. A B [2]  A  B
			[Total: 13]

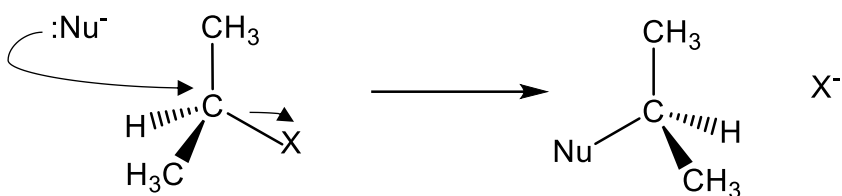
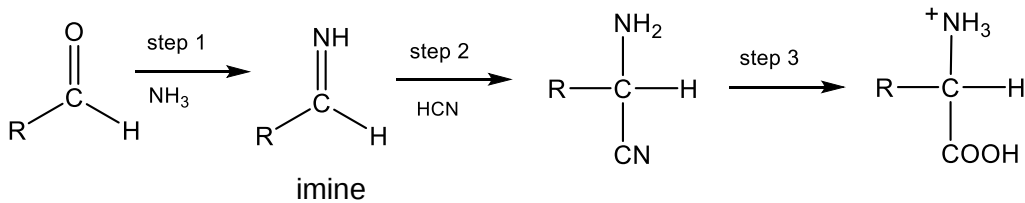
2	Chromium is a transition metal that is valued for its high resistance to corrosion and is added to steel to form stainless steel. While chromium can exist in various oxidation states, the most common oxidation state is the +3 state.
(a)	Hydrated chromium (III) chloride exists as isomers, with the general formula of $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$. One such isomer is $[\text{CrCl}_2(\text{H}_2\text{O})_4]\text{Cl} \cdot 2\text{H}_2\text{O}$ and it appears dark green. Suggest the formula of two other isomers of hydrated chromium (III) chloride. isomer 1: isomer 2:[2]
	$[\text{CrCl}_3(\text{H}_2\text{O})_3] \cdot 3\text{H}_2\text{O}$ $[\text{CrCl}(\text{H}_2\text{O})_5]\text{Cl}_2 \cdot \text{H}_2\text{O}$ $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$
(b)	When a sample of hydrated chromium (III) chloride is added to excess water, a green solution of $\text{CrCl}_3(\text{aq})$ is obtained. Fig. 2.1 shows the reactions that aqueous CrCl_3 can undergo. <pre> graph TD A["CrCl3 (aq) green solution"] -- "Na2CO3 (aq)" --> B["grey-green precipitate A + effervescence of gas B"] A -- "left to stand" --> C["D violet solution"] B -- "excess NaOH (aq)" --> D["C dark green solution"] D -- "H2O2 (aq), heat" --> E["CrO4^2- (aq) yellow solution"] </pre> Fig. 2.1
(i)	Identify precipitate A and gas B. precipitate A:

			gas B:[2] A: $\text{Cr}(\text{OH})_3$ or $\text{Cr}(\text{OH})_3(\text{H}_2\text{O})_3$ or chromium(III) hydroxide, B: CO_2 or carbon dioxide
		(ii)	Suggest the formula of compound C. [1] $\text{Na}_3[\text{Cr}(\text{OH})_6]$
		(iii)	Explain why solutions of transition metal compounds are often coloured. The d orbitals are partially filled. In the presence of ligands, the degenerate d-orbitals/d-subshell are split into two groups of different energy levels. An electron in the lower energy level orbital can absorb energy equivalent to the energy gap which corresponds to a certain wavelength of light from the visible region of the electromagnetic spectrum and be promoted to the higher energy level orbital. The colour seen is complementary to the colour absorbed.
		(iv)	Given that the oxidation state of chromium in compound D is still +3, suggest what happened that caused the colour change from green to violet. Ligand exchange occurred. The energy gap between lower energy level d orbitals and the higher energy level d orbitals changed and hence a different wavelength of visible light is absorbed.
		(v)	State the role of H_2O_2 (aq). [1] oxidising agent
			Comments: - Students need to recognise that the oxidation state increased from +3 to +6 and hence H_2O_2 acted as an oxidising agent. - This question was well answered.
		(vi)	Write a balanced equation for the formation of CrO_4^{2-} from C. [1] $2[(\text{Cr}(\text{OH})_6)]^{3-} + 3\text{H}_2\text{O}_2 \rightarrow 2\text{CrO}_4^{2-} + 2\text{OH}^- + 8\text{H}_2\text{O}$ OR

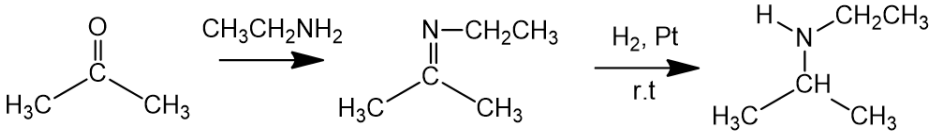
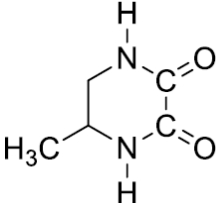
			$2\text{Na}_3[\text{Cr}(\text{OH})_6] + 3\text{H}_2\text{O}_2 \rightarrow 2\text{Na}_2\text{CrO}_4 + 2\text{NaOH} + 8\text{H}_2\text{O}$																								
(c)	Fig. 2.2 shows the first seven ionisation energies of calcium and chromium. <table border="1"><caption>Approximate data points from Fig. 2.2</caption><thead><tr><th>Number of electrons removed</th><th>Graph X (Ca) IE</th><th>Graph Y (Cr) IE</th></tr></thead><tbody><tr><td>1</td><td>Low</td><td>Low</td></tr><tr><td>2</td><td>Low</td><td>Low</td></tr><tr><td>3</td><td>Medium</td><td>Medium-High</td></tr><tr><td>4</td><td>Medium</td><td>Medium</td></tr><tr><td>5</td><td>Medium-High</td><td>Medium-High</td></tr><tr><td>6</td><td>Very High</td><td>Medium</td></tr><tr><td>7</td><td>Very High</td><td>Very High</td></tr></tbody></table> Fig. 2.2 State and explain which graph shows the successive ionisation energies for calcium.			Number of electrons removed	Graph X (Ca) IE	Graph Y (Cr) IE	1	Low	Low	2	Low	Low	3	Medium	Medium-High	4	Medium	Medium	5	Medium-High	Medium-High	6	Very High	Medium	7	Very High	Very High
Number of electrons removed	Graph X (Ca) IE	Graph Y (Cr) IE																									
1	Low	Low																									
2	Low	Low																									
3	Medium	Medium-High																									
4	Medium	Medium																									
5	Medium-High	Medium-High																									
6	Very High	Medium																									
7	Very High	Very High																									
	Graph Y showed the variation in successive ionisation energies (IE) for Ca. This is because there is a large jump between the 2nd and 3rd IE showing that the 3rd electron is removed from an inner shell and the atom has 2 valence electrons.																										
			[Total: 14]																								

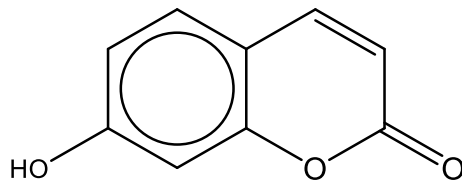
3	(a)	The variation of the volume with temperature of a fixed mass of an ideal gas at constant pressure may be represented by a relationship known as Charles's law, $V = k T$ where V is the volume of a gas, T is the temperature in Kelvin and k is a constant. An experiment was carried out to attempt to verify this law using a gas, phosphine, PH_3. The experiment was repeated several times at different temperatures. Fig. 3.1 was plotted to determine the relationship between the volume of the gas and its temperature. <table><caption>Data points from Fig. 3.1</caption><thead><tr><th>Temperature of the gas / K</th><th>Volume of the gas / cm³</th></tr></thead><tbody><tr><td>21</td><td>20</td></tr><tr><td>27</td><td>25</td></tr><tr><td>31</td><td>29</td></tr><tr><td>37</td><td>35</td></tr><tr><td>43</td><td>40</td></tr></tbody></table> Fig. 3.1 temperature of the gas/K	Temperature of the gas / K	Volume of the gas / cm³	21	20	27	25	31	29	37	35	43	40
Temperature of the gas / K	Volume of the gas / cm³													
21	20													
27	25													
31	29													
37	35													
43	40													
	(i)	Based on Fig. 3.1, state whether the result is consistent with Charles's law. Give a reason for your answer.												
		Charles's law is confirmed because a straight line passing through the origin with a positive gradient (direct proportionality) is obtained.												
	(ii)	State the two properties of a gas necessary for it to approach ideal behaviour.												

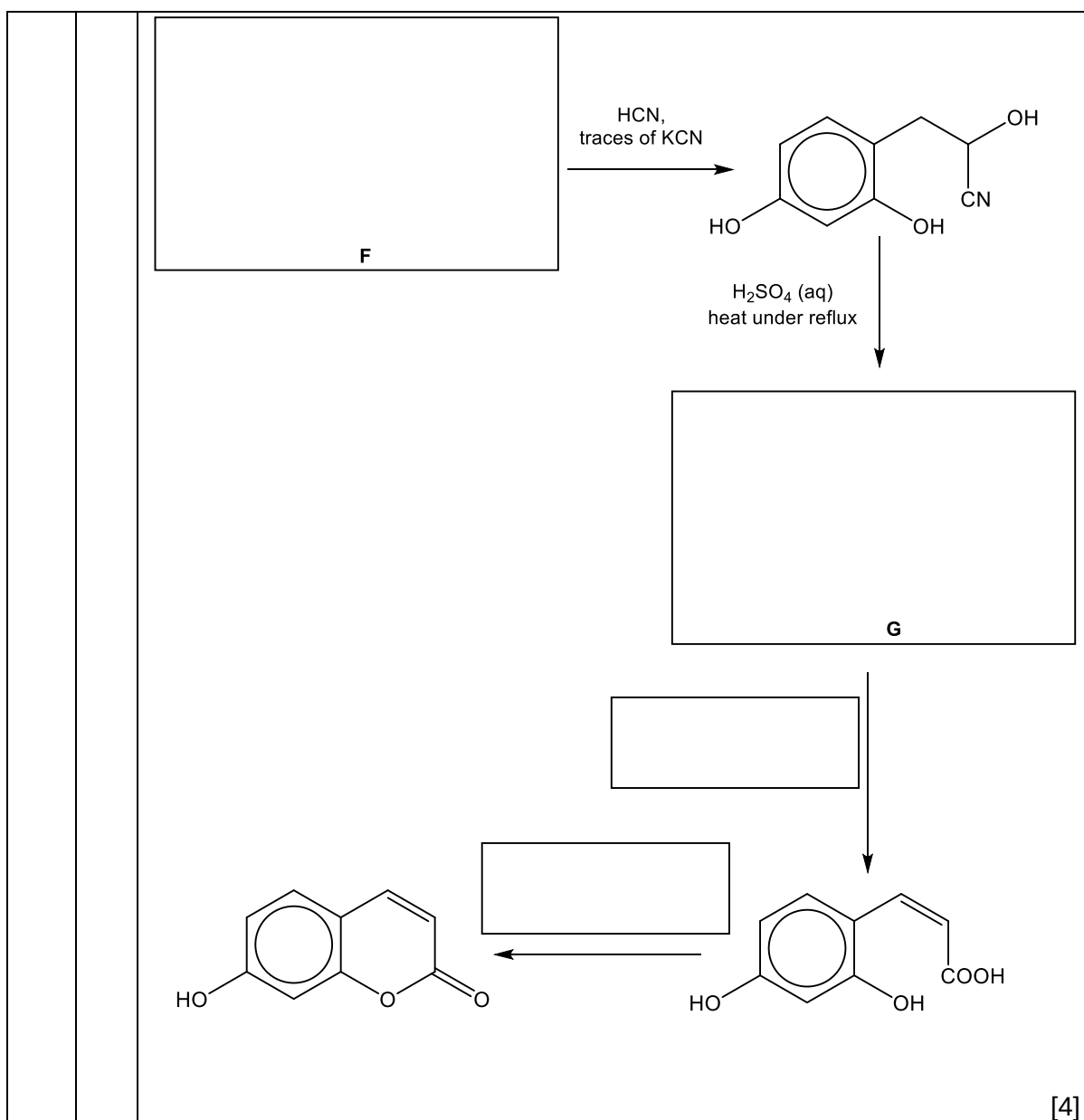
			The gas must have negligible intermolecular forces of attraction. There is negligible size/volume of the gaseous molecules/particles compared to the size of the container/volume of gas.
	(b)	Orange street lamps contain sodium with a small amount of neon. The light is produced when gaseous atoms are ionised in an electric field. When it is first turned on, the lamps emit a red glow characteristic of neon, but after some time, the orange glow of sodium predominates. Explain the order for the appearance of the colours. Use relevant data from the <i>Data Booklet</i> in your answer. [2]	
		The ionisation energy of Na is 494 kJmol^{-1} which is smaller than Ne which has an ionisation energy of 2080 kJmol^{-1}. However, Na is a solid at room temperature needs to be vapourised to Na(g) before ionization. OR Ne is already a gas at room temperature can be ionised immediately. Since time is required to vapourise the Na prior to ionisation, red glow of Ne is observed before the orange glow of sodium.	
	(c)	4-chloromethylbenzene and (chloromethyl)benzene are isomers.  4-chloromethylbenzene  (chloromethyl)benzene	
		(i)	Explain the difference in their reactivities towards hot aqueous sodium hydroxide.
			(Chloromethyl)benzene undergoes nucleophilic substitution with hot NaOH(aq) readily while 4-chloromethylbenzene requires vigorous conditions (or does not react). OR Chloromethylbenzene is more reactive than 4-chloromethylbenzene (or any other form of comparison of reactivity). The p-orbital of Cl in 4-chloromethylbenzene can overlap with the π electron cloud/p orbitals of the benzene ring, thus lone pairs of electrons of Cl delocalizes into benzene ring resulting in partial double bond between C and Cl and strengthening it. Hence, the C-Cl bond does not undergo nucleophilic substitution with NaOH(aq) easily.
		(ii)	Alkyl halides can undergo nucleophilic substitution, with the simplified general mechanism as shown below:

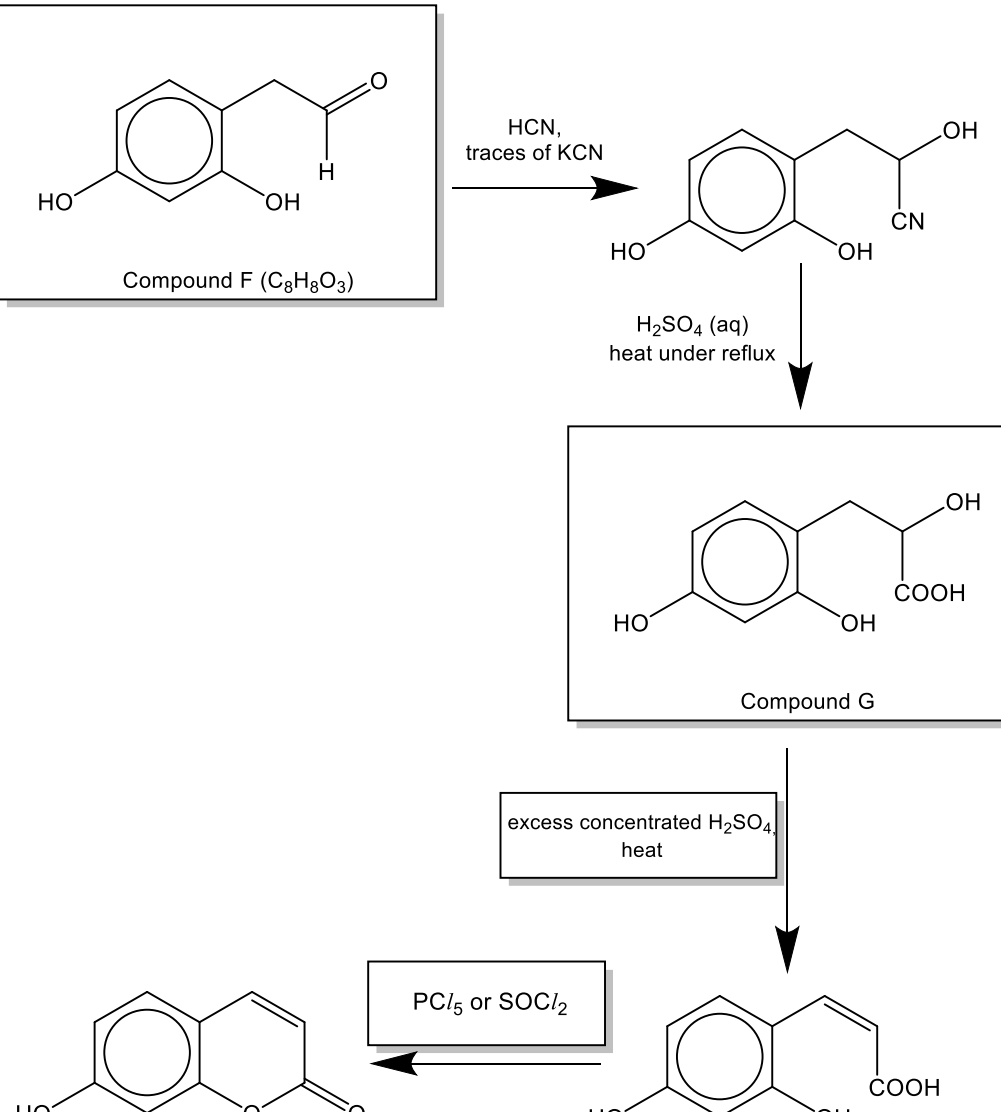
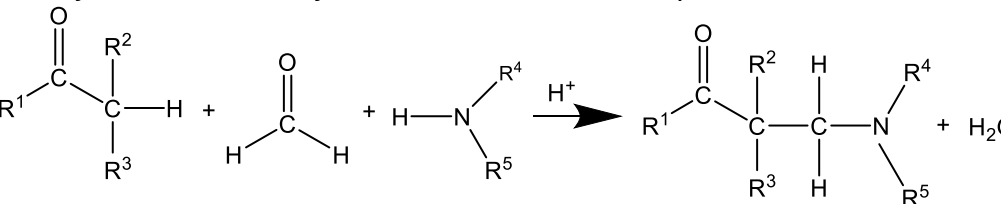
		 The mechanism involves the heterolytic fission of the C–X bond, resulting in X leaving with a pair of electrons as X[–]. X[–] is called the leaving group. When alkyl halide undergoes nucleophilic substitution, one factor that affects its reactivity is the basicity of X[–]. In general, when X[–] is a stronger base, it is a poorer leaving group. This results in the alkyl halide being less reactive towards nucleophilic substitution. Using the data given in Table 3.1 and relevant data from the <i>Data Booklet</i>, suggest two reasons to explain why (iodomethyl)benzene reacts much faster than (chloromethyl)benzene. <table data-bbox="649 748 1152 904"><caption>Table 3.1</caption><tr><th>halide</th><th>pK_b</th></tr><tr><td>Cl[–]</td><td>21.0</td></tr><tr><td>I[–]</td><td>24.0</td></tr></table>	halide	pK _b	Cl [–]	21.0	I [–]	24.0
halide	pK _b							
Cl [–]	21.0							
I [–]	24.0							
		(Iodomethyl)benzene reacts much faster than (chloromethyl)benzene due to the weaker C–X bond and more stable X[–] formed. BE(C–I) 340 kJ mol^{–1} is smaller than BE(C–Cl) 240 kJ mol^{–1}. C–I bond is weaker than C–Cl, hence (iodomethyl)benzene tends to react much faster than (chloromethyl)benzene. From the larger pK_b (or smaller K_b), I[–] is a weaker base making it a good leaving group, I[–] is a more stable conjugate base. Hence, I[–] is formed more readily and hence (iodomethyl)benzene tends to react much faster than (chloromethyl)benzene.						
	(d)	The first known synthesis of an amino acid occurred in 1850 in the laboratory of Adolf Strecker. 						
	(i)	Name the type of reaction in step 1. [1] Condensation						
		Comments: This reaction resembles <u>condensation</u> reaction between carbonyl and 2,4 DNPH.						

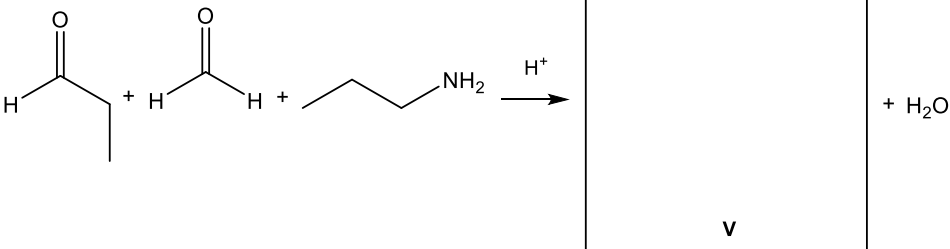
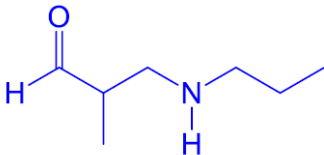
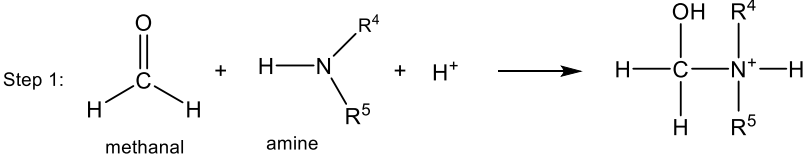
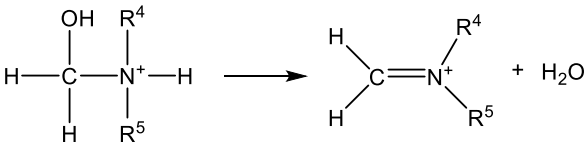
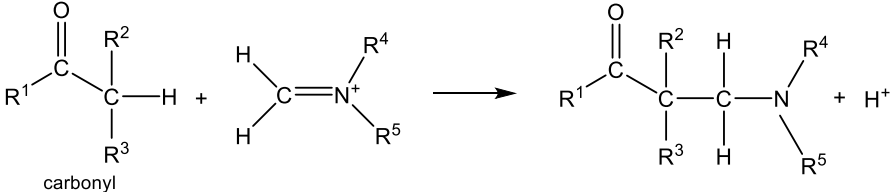
		(ii) In step 2, the reaction proceeds via two stages: (I) acid-base reaction between the N atom in imine and HCN (II) nucleophilic attack on C atom by CN^- Draw the step 2 for this mechanism. Show all relevant charges, dipoles, lone pairs and curly arrows. [2]
		All steps must have appropriate lone pairs, arrows, charges and must be balanced.
		Comments: Dipoles should not be drawn for polar bonds in polyatomic ions as the electron distribution within the ion may not be the same as that depicted by the polarity of the bond.
		(iii) An imine intermediate is also formed in preparing secondary amines from ketones. imine Suggest the synthetic route by giving all the reagents, conditions and intermediate for the preparation of $(\text{CH}_3)_2\text{CHNH}(\text{CH}_2\text{CH}_3)$. [4]

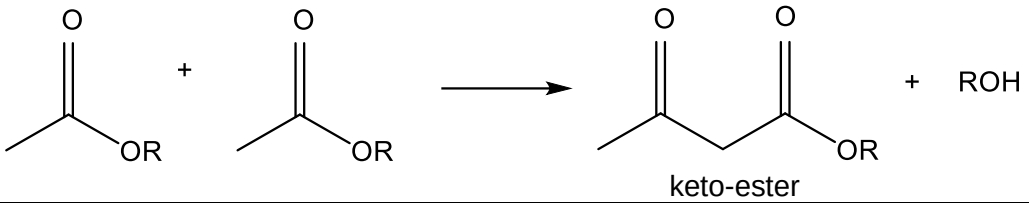
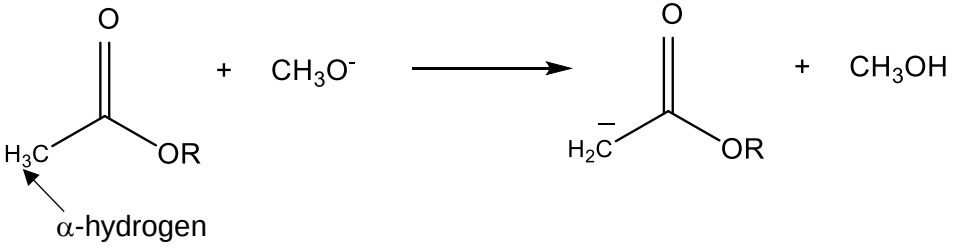
			 Alternative Ans: LiAlH₄ in dry ether, r.t. as reducing agent in second step
		(iv)	Another amine X, CH₃CH(NH₂)CH₂NH₂ reacts with ethanedioyl dichloride (COCl)₂ to produce compound Y with molecular formula C₅H₈N₂O₂. Suggest the structure for compound Y. [1]
			
			[Total: 16]

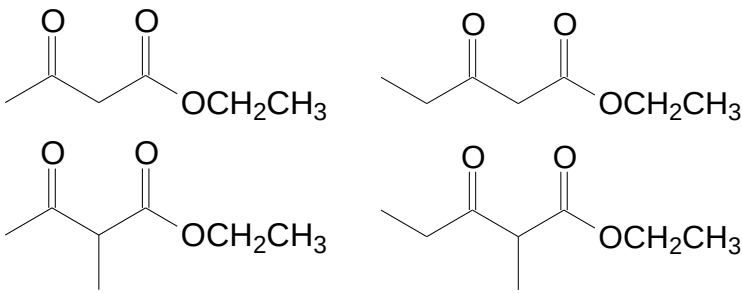
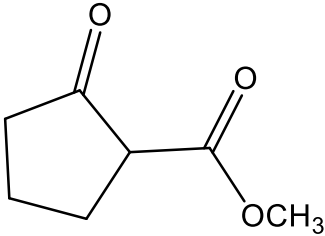
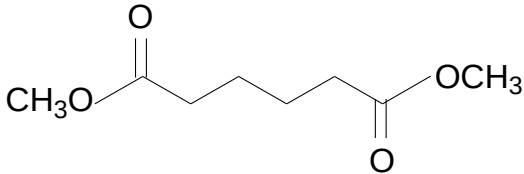
4	(a)	Umbelliferone is a yellowish-white crystalline solid occurring naturally in carrots and is used as a sunscreen agent. It has the structure shown below:  The following reaction scheme shows how umbelliferone may be synthesised from compound F, which has a molecular formula of C₈H₈O₃. In the boxes provided, fill in the missing • reagents and conditions, • structures of compounds F and G.
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		 Compound F ($C_8H_8O_3$) HCN, traces of KCN H_2SO_4 (aq) heat under reflux Compound G excess concentrated H_2SO_4 heat $POCl_3$ or $SOCl_2$
	(b)	Mannich reaction is an organic reaction which consists of an amino alkylation of the hydrogen atom adjacent to a carbonyl functional group, by methanal and a primary amine, secondary amine or ammonia, in the presence of acid.  Carbonyl Methanal Amine
	(i)	Predict the structure of V for each of the following Mannich reactions.

			 + H₂O v [1]
			 [1] V (C₇H₁₅NO)
		(ii)	The proposed mechanism for Mannich reaction is as follows: Step 1:  methanal amine  + H₂O  carbonyl Suggest the types of reaction that occurred in Step 1 and 2. Step 1..... Step 2 [2] Step 1: Nucleophilic addition Step 2: Elimination

	(c)	The Claisen condensation of esters involves the formation of a new carbon-carbon bond between two esters. The products are a keto-ester and an alcohol. The reaction takes place in the presence of a strong base e.g. CH_3O^-. 
	(i)	The first step in Claisen condensation involves the strong base CH_3O^- removing an α-hydrogen atom (i.e. the hydrogen atom bonded to the carbon atom next to the carbonyl carbon atom) in an acid-base reaction as shown below.  Suggest a reason why the α-hydrogen atom of the ester is acidic. [1]
		The conjugate base of the ester is stabilised as the negative charge can be dispersed by delocalisation into the adjacent $\text{C}=\text{O}$ group.
	(ii)	With reference to the acid-base reaction mentioned in (c)(i), suggest why each of the following reactions does not take place in the presence of CH_3O^-. (1) $\text{CH}_3\text{CH}_3 \longrightarrow ^-\text{CH}_2\text{CH}_3$ (2) $\text{CH}_3\text{COOH} \longrightarrow ^-\text{CH}_2\text{COOH}$ [2]
		(1) The conjugate base of ethane is unstable as the electron-donating methyl group will intensify the negative charge. OR CH_3O^- is a more stable base than CH_3CH_2^- as the negative charge is stabilized by electronegative O atom in CH_3O^- (2) The conjugate base formed $-\text{COO}^-$ in deprotonating $-\text{COOH}$ is more stable as negative charge is delocalised over 2 electronegative O atoms. OR O is more electronegative than C, hence stabilised negative charge more.

		(iii)	Draw two possible structures of all the keto-esters formed when a mixture of $\text{CH}_3\text{CO}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}_3$ undergo the Claisen condensation. [2]
			
		(iv)	Draw the structure of the reactant that can be used to prepare the following compound by the Claisen condensation.  [1]
			
			[Total: 13]

- 5 Hydrogen fuel cells use hydrogen gas with oxygen gas to form water and generate electricity. Two types of hydrogen fuel cells have been used to power vehicles. **Alkaline fuel cells (AFCs)** use an aqueous solution of potassium hydroxide as the electrolyte. In recent years, novel AFCs that use an alkaline polymer membrane as the electrolyte have been developed. Fig. 5.1 shows the diagram of an AFC.

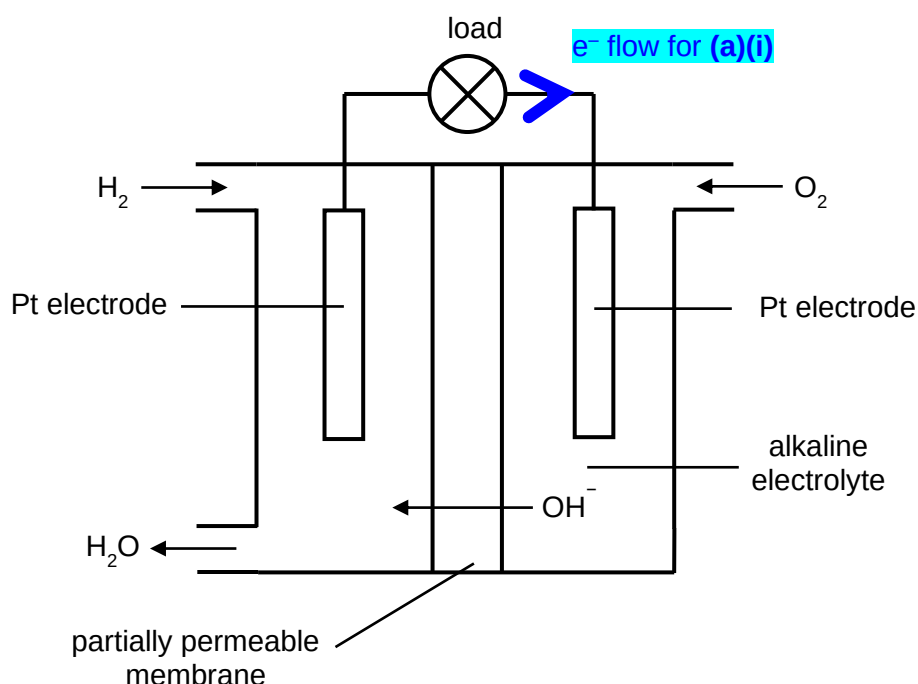
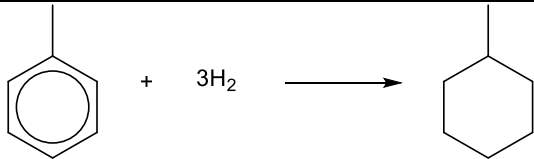


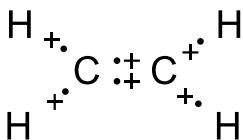
Fig. 5.1

Proton exchange membrane (PEM) fuel cells use a solid polymer as an electrolyte and porous carbon electrodes containing a platinum or platinum alloy catalyst. It has the same overall reaction as the AFC but uses an acidic electrolyte.

(a)	(i)	On Fig. 5.1, draw the direction of electron flow. [1]
	(ii)	When adjusted for standard conditions, an AFC is known to generate 1.23 V of electromotive force. With reference to the <i>Data Booklet</i> , write the half equation for each electrode and the overall equation. [2]
		anode: ... $H_2 + 2OH^- \rightarrow 2H_2O + 2e^-$ cathode: ... $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$ overall: ... $2H_2 + O_2 \rightarrow 2H_2O$ [2]
	(iii)	Using information from (a)(ii), calculate the Gibbs free energy change per mol of water generated in the AFC. [1]
		$\Delta G = -nFE_{\text{cell}} = -(2)(96500)(1.23)$ $= -237390 \text{ J mol}^{-1} = -237 \text{ kJ mol}^{-1} \text{ (3 s.f.)}$

		(iv)	Calculate the $E_{cell}^{\ominus}$ of the PEM fuel cell. [2]												
			$E_{cell}^{\ominus} = E_{cathode}^{\ominus} - E_{anode}^{\ominus}$ $= 1.23 - 0.00$ $= +1.23 \text{ V}$												
		(v)	Comment on the voltages generated by the AFC and PEM fuel cell. [1]												
			They have the same voltage because they undergo the same overall chemical reaction, which possesses the same chemical energy.												
	(b)	The disadvantage of transporting hydrogen gas is one of the factors limiting the use of hydrogen fuel cells. More recently, liquid organic hydrogen carriers (LOHC) such as the methylbenzene / methylcyclohexane system have been used to transport hydrogen safely over long distances. At its source, hydrogen is 'loaded' on the LOHC by reacting hydrogen with methylbenzene in the presence of platinum catalyst to produce methylcyclohexane. Hydrogen is 'unloaded' from the LOHC by reacting methylcyclohexane with the catalyst to regenerate methylbenzene. Table 5.1 contains information about methylbenzene and methylcyclohexane. Table 5.1 <table border="1" style="margin-left: auto; margin-right: auto;"> <thead> <tr> <th>liquids</th><th>density under conditions for transportation / kg m⁻³</th><th>boiling point at 1 atm / °C</th><th>flammability</th></tr> </thead> <tbody> <tr> <td>hydrogen</td><td>71</td><td>-252.9</td><td>very high</td></tr> <tr> <td>methylcyclohexane</td><td>770</td><td>101</td><td>high</td></tr> </tbody> </table>		liquids	density under conditions for transportation / kg m ⁻³	boiling point at 1 atm / °C	flammability	hydrogen	71	-252.9	very high	methylcyclohexane	770	101	high
liquids	density under conditions for transportation / kg m ⁻³	boiling point at 1 atm / °C	flammability												
hydrogen	71	-252.9	very high												
methylcyclohexane	770	101	high												
		(i)	Write a balanced equation for the loading of hydrogen on the LOHC.												
															
		(ii)	State the type of catalyst used in the loading of the LOHC. Describe how it catalyses the reaction. [3]												
			Heterogeneous catalyst. - Hydrogen gas and methylbenzene adsorb on the surface of the platinum catalyst by forming weak bonds with the catalyst. - Intramolecular covalent bonds are weakened, hence E_a is lowered when the reactants bond to the catalyst, allowing the reaction to occur more easily. - Reactant molecules are brought into close contact at the catalyst surface. The surface concentration of the reactants increases, so the rate of reaction increases.												

			Methylcyclohexane desorb from the surface of the catalyst, allowing more reactants to occupy the site.
		(iii)	Calculate the mass of H ₂ transported per m ³ of methylcyclohexane. [2]
			Amount of methylcyclohexane per m³ $= 770\,000\text{ g m}^{-3} \div 98\text{ g mol}^{-1} = 7857\text{ mol}$ Amount of hydrogen per m³ of methylcyclohexane $= 7857 \times 3 = 23571\text{ mol}$ Mass of hydrogen per m³ of methylcyclohexane $= 23571\text{ mol m}^{-3} \times 2\text{ g mol}^{-1}$ $= 47142\text{ g m}^{-3} = 47.1\text{ kg m}^{-3}$ (3 s.f.)
		(iv)	Suggest whether it is advantageous for the transport of hydrogen to be changed from liquid hydrogen to the use of LOHC. [1]
			This change is advantageous as LOHC is less flammable / less explosive than hydrogen. OR This change is advantageous because LOHC has a higher boiling point / it is in the liquid state at room conditions, so it can be transported without cooling OR high pressure unlike H₂. OR This change is not advantageous as LOHC has a lower mass (or less) of hydrogen gas transported per unit volume.
	(c)		The hydrogen supply for fuel cells have been classified according to its impact on the environment. Green hydrogen refers to the gas that is produced using renewable energy, without greenhouse gas emissions. For example, electrolysis of water powered by solar, wind or geothermal energy produces green hydrogen.
		(i)	Currently the most common method for generating hydrogen gas involves two successive reactions: 1) steam-methane reformation 2) water-gas shift reaction Using this method, methane reacts with steam to form four times the amount of hydrogen gas along with a side product. Write the overall balanced equation for the above method. [1]
			$\text{CH}_4 + 2\text{H}_2\text{O} \rightarrow 4\text{H}_2 + \text{CO}_2$
		(ii)	Hence suggest whether the method in (c)(i) is a source of green hydrogen. [1]
			It is not a source of green hydrogen as the greenhouse gas CO ₂ is formed.

		(iii)	Hydrogen produced by steam-methane reformation may contain trace contaminants such as ethene. Draw the dot-and-cross diagram of ethene. State the bond angle and shape around the carbon atom.
			Bond angle: 120° Shape: Trigonal planar  [2]
		(d)	Electric vehicles (EVs) have been known as a greener form of transportation than petrol powered vehicles because greenhouse gases are not emitted from EVs. By considering the source of the electrical energy, comment on the validity of this claim.
			 [1] The claim is valid because using renewable sources of energy such as solar, wind or geothermal energy to generate electricity does not release greenhouse gases. OR The claim is not valid since the burning of fossil fuels / fuels / coal to generate electricity releases greenhouse gases. Further information about electric vehicles: • Watch CNA Insider 'Switching To Electric Car: Is It Worth It?' https://youtu.be/cc9iYjPLnFM?t=693
			[Total: 19]